fourth-generation UAVs and manned–unmanned collaborative ecosystems, integrating artificial intelligence (AI), stealth, swarm technology, and digital cloud-based warfare into a unified operational network (Vogel, 2021).

At its core, the FCAS is designed as a System of Systems (SoS), a multi-layered network of manned fighters, unmanned aerial vehicles (UAVs) known as Remote Carriers (RCs), and space and cyber assets all operating under a combat cloud architecture that allows for real-time information sharing and coordinated mission execution. The Next Generation Fighter (NGF) serves as the central manned aircraft, surrounded by autonomous RC drones that perform reconnaissance, electronic warfare, or precision strike missions under AI supervision.

Key innovations of the FCAS program include:

- *AI-driven combat cloud*: A secure, distributed digital ecosystem that fuses sensor data from all platforms, enabling shared situational awareness and AI-assisted decision-making.
- *Swarm and teaming capability*: RC drones can operate in autonomous swarms, supporting the NGF with offensive or defensive maneuvers, creating a force-multiplying effect on the battlefield.
- *Stealth and survivability*: The FCAS UAVs and NGF feature low-observable materials, adaptive aerodynamics, and electromagnetic shielding, allowing deep-strike capabilities in contested airspace.
- *Multi-domain interoperability*: The system integrates air, land, sea, space, and cyber domains, establishing a true network-centric combat environment across European defense forces.

The Remote Carriers, acting as loyal wingmen, are central to the FCAS ecosystem. They range from small, expendable drones for electronic deception or sensor extension, to larger, reusable platforms for strategic